

Lesson-1: Quiz (graded)

1

Fill in the Blank 1 point

Which of the following statements about Network Science are **TRUE**?

It is a new branch of graph theory that focuses on mathematical properties of non-trivial graphs

It studies complex systems using tools from graph theory and other disciplines

It is a top-down way to understand complex systems based on their architecture

It focuses on data and measurements that describe the system architecture as a network (or graph)

⋮ FALSE

⋮ TRUE

⋮ FALSE

⋮ TRUE

⋮ FALSE

⋮ TRUE

⋮ FALSE

⋮ TRUE

2

Multiple answer 1 point

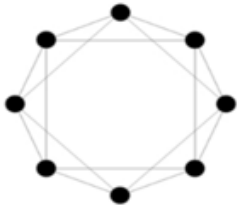
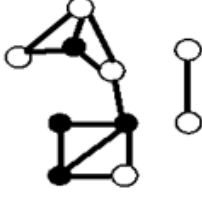
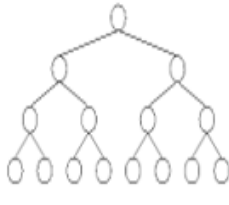
Which of the following topics are covered in this Network Science course (select all)?

- ☐ Identifying network communities
- ☐ Dynamics of networks (how networks change over time)
- ☐ Dynamics on networks (dynamic processes running on networks)
- ☐ Identifying the "centrality" of nodes and edges

3

Multiple choice 1 point

Which of the following networks can be categorized as “non-trivial” graphs (at least according to Lesson-1)?

A	B	C	D
A random network in which any two nodes are connected with the same probability			

- ☐ A
- ☐ B
- ☐ C
- ☐ D

4

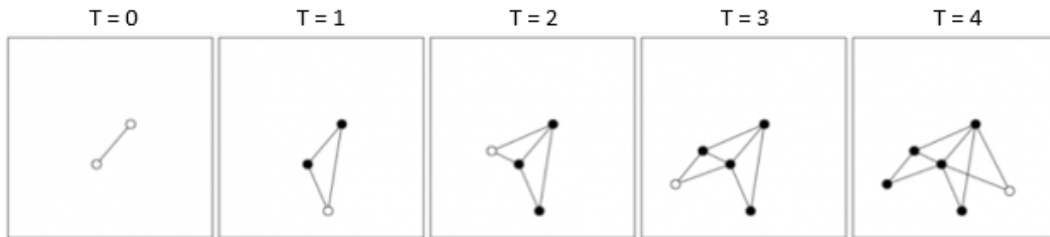
Multiple choice 1 point

What is the main difference between Network Science and Graph Theory?

- ☐ Network Science focuses on graphs that represent the architecture of real-world complex systems, while Graph Theory focuses on abstract mathematical topologies
- ☐ Network Science is only empirical (no mathematical analysis) while Graph Theory is a mathematical discipline
- ☐ Network Science only focuses on large graphs
- ☐ None of the above

5

Fill in the Blank 1 point



This sequence shows the first 5 steps of a growing network. White vertices refer to the newly added vertex at each timestamp. Which of the following statements about this graph are **TRUE**?

This is an example of a regular network growing over time

This is an example of a dynamic process taking place on a static network

This is an example of a process in which the network is dynamic

New nodes are directly connected to the original two nodes at all time steps (the two nodes at $T=0$)

☐ FALSE

☐ TRUE

☐ FALSE

☐ TRUE

☐ FALSE

☐ TRUE

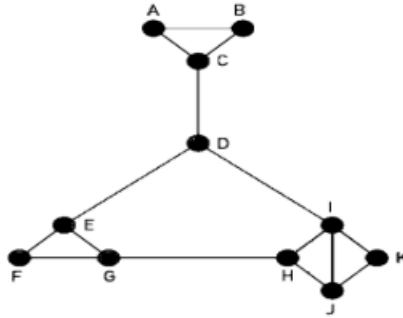
☐ FALSE

☐ TRUE

6

Fill in the Blank 1 point

Which of the following statement(s) about this graph are **TRUE**? (Select all)



The highest degree node is D (degree is the number of connections of a node)

The maximum (shortest path) distance between any two nodes is 5 hops (links)

The betweenness centrality (i.e., number of shortest paths, between all node pairs, that traverse a node) of node D is higher than of node A

H and J are in the same community

☐ FALSE

☐ TRUE

☐ FALSE

☐ TRUE

☐ FALSE

☐ TRUE

☐ FALSE

☐ TRUE

7

Multiple choice 1 point

Which of the following can be example(s) of a scale-free network?

- ☐ A diamond crystal structure in which a pattern of atoms is repeated regularly in space
- ☐ Any random network
- ☐ Any complex network
- ☐ A social network in which few people have a much greater number of friends than most other people